Hyperreal-valued measures on the circle

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This short note was inspired by [3], which gives a simple proof of the fact that for any real-valued probability measure there is a regular,[1] hyperreal-valued one which approximates it up to an infinitesimal. Here we give a similarly simple proof of a well-known result of Bernstein and Wattenberg [1] which asserts the existence of a regular, hyperreal-valued probability measure on all subsets of $[0, 1)$ which is infinitely close to Lebesgue measure (on measurable sets), uniform on points, and translation-invariant up to an infinitesimal.

Preliminaries: Let $\Omega = \mathbb{R}/\mathbb{Z}$, identified with the interval $[0, 1)$ with addition mod 1. For $X \subseteq \Omega$ and $r \in \Omega$, we write $X + r$ for the set $\{x + r : x \in X\}$. We also write $\mathcal{L}$ for the σ-algebra of Lebesgue-measurable subsets of Ω and use λ to denote Lebesgue measure.

Where x and y are members of any ordered field ${}^*\mathbb{R}$ extending $\mathbb{R}$, by $x \simeq y$ we mean that $|x - y| < 1/n$ for all $n \geq 1$ (informally: x and y are "infinitely close"). And, as usual, $\mathfrak{A} \succ \mathfrak{B}$ means that $\mathfrak{A}$ is an elementary extension of $\mathfrak{B}$.

Theorem (Bernstein–Wattenberg)**.** *There exist a field* ${}^*\mathbb{R} \succ \mathbb{R}$ *and a finitely additive* $\mu\colon \mathcal{P}(\Omega) \to {}^*\mathbb{R}$ *with* $\mu(\Omega) = 1$*, such that*

(i) μ *is regular, i.e.,* $\mu(X) > 0$ *for all nonempty* $X \subseteq \Omega$*;*

(ii) $\mu(X) \simeq \lambda(X)$ *for every* $X \in \mathcal{L}$*;*

(iii) μ *is uniform on points, i.e.,* $\mu(\{x\}) = \mu(\{y\})$ *for all* $x, y \in \Omega$*; and*

(iv) μ *is translation-invariant up to an infinitesimal, i.e.,* $\mu(X+r) \simeq \mu(X)$ *for every* $X \subseteq \Omega$ *and* $r \in \Omega$*.*

Two quick remarks before we embark upon a proof.

[1] We follow the literature on hyperreal-valued probability in calling a measure *regular* iff it assigns positive measure to every nonempty set in its domain; this is unrelated to the measure-theoretic notion of inner/outer regularity.

Observation 1. Condition *(iv)* can't be strengthened to *exact* translation-invariance. Let $A = \{kx : k \geq 1\}$ for some irrational $x \in \Omega$; the kx's are pairwise distinct, so $A + x = A \setminus \{x\}$, and therefore exact invariance would imply $\mu(A) = \mu(A + x) + \mu(\{x\}) > \mu(A)$: contradiction.

Observation 2. μ is not countably additive. Take distinct points $x_1, x_2, \ldots$ in Ω, with $\mu(\{x_i\}) = \varepsilon > 0$; countable additivity requires the partial sums $(n\varepsilon)_{n\geq 1}$ to converge, which is impossible since no two of them can lie within $\varepsilon/2$ of any limit.

Now to our proof of the theorem. The basic idea, as in [3], is to apply the compactness theorem to a suitable finitely satisfiable set of statements Γ. We work in the signature $(0, 1, +, \cdot, <)$ of ordered fields, augmented with a constant $\dot{r}$ for each $r \in \mathbb{R}$ and a constant $\dot{\mu}(X)$ for each $X \subseteq \Omega$, and take Γ to consist of:

(a) the elementary diagram of $\mathbb{R}$, with $\dot{r}$ interpreted as r;

(b) "$\dot{\mu}(\Omega) = 1$";

(c) "$\dot{\mu}(X) > 0$", for each nonempty $X \subseteq \Omega$;

(d) "$\dot{\mu}(X \cup Y) = \dot{\mu}(X) + \dot{\mu}(Y)$", for each pair of disjoint $X, Y \subseteq \Omega$;

(e) "$|\dot{\mu}(X) - \dot{x}| < \dot{q}$", for each $X \in \mathcal{L}$ with $\lambda(X) = x$, and for every $q = \frac{1}{n}$ with $n \geq 1$;

(f) "$|\dot{\mu}(X + r) - \dot{\mu}(X)| < \dot{q}$", for each $X \subseteq \Omega$, $r \in \Omega$, and for every $q = \frac{1}{n}$ with $n \geq 1$; and

(g) "$\dot{\mu}(\{x\}) = \dot{\mu}(\{y\})$", for each $x, y \in \Omega$.

Any model ${}^*\mathbb{R}$ of Γ will be an elementary extension of $\mathbb{R}$, by *(a)*, and will contain values $\mu(X) \coloneqq \dot{\mu}(X)^{{}^*\mathbb{R}}$ satisfying conditions *(i)–(iv)*, by *(b)–(g)*. As usual, all the spadework goes into proving that every finite $\Gamma_0 \subseteq \Gamma$ has a model, to which task we now turn.

First, we extract a useful lemma from the main argument of [3].

Lemma 1. *Let $\mathcal{F}$ be an algebra of subsets of a set Ω, and let $\nu \colon \mathcal{F} \to [0, 1]$ be finitely additive with $\nu(\Omega) = 1$. Given any finite Boolean algebra $\mathcal{B}$ of subsets of Ω and "tolerance" $\delta > 0$, there is a finitely additive $\rho \colon \mathcal{B} \to \mathbb{R}$ such that*

(i) *$\rho(\Omega) = 1$, and ρ is regular;*

(ii) $|\rho(X) - \nu(X)| < \delta$ *for every* $X \in \mathcal{B} \cap \mathcal{F}$*; and*

(iii) $\rho(\{x\}) = \rho(\{y\})$ *whenever* $\{x\}, \{y\} \in \mathcal{B} \cap \mathcal{F}$ *and* $\nu(\{x\}) = \nu(\{y\}) = 0$.

To keep this note self-contained, we'll reprove this lemma for ourselves here.

Proof. Since $\mathcal{F}$ is an algebra, $\mathcal{B} \cap \mathcal{F}$ is a finite subalgebra of $\mathcal{B}$. Let $\mathcal{C} = \mathrm{At}(\mathcal{B} \cap \mathcal{F})$ be the set of atoms of this subalgebra; note that this is a partition of Ω. Every atom A of $\mathcal{B}$ lies in precisely one $C \in \mathcal{C}$, since each such C is itself in $\mathcal{B}$ and, by the fact that A is an atom, either $A \subseteq C$ or $A \cap C = \emptyset$; and every $X \in \mathcal{B} \cap \mathcal{F}$ is a union of the members of $\mathcal{C}$ which it contains.[2]

Now partition $\mathcal{C}$ into $\mathcal{C}_0 = \{C : \nu(C) = 0\}$ and $\mathcal{C}_+ = \{C : \nu(C) > 0\}$, with respective cardinalities c_0 and c_+, say. We fix ε with $0 < \varepsilon < \min(\{\delta\} \cup \{\nu(C) : C \in \mathcal{C}_+\})$, and put $\eta = \varepsilon$ if $c_0 > 0$ and $\eta = 0$ if $c_0 = 0$. Define

$$\rho(C) = \begin{cases} \frac{\eta}{c_0} & C \in \mathcal{C}_0, \\ \nu(C) - \frac{\eta}{c_+} & C \in \mathcal{C}_+, \end{cases}$$

where we understand $\frac{\eta}{c_0}$ to be 0 if $c_0 = 0$.

For each $C \in \mathcal{C}$ let n_C be the number of atoms of $\mathcal{B}$ contained in C, and for each such atom A set $\rho(A) = \frac{\rho(C)}{n_C}$. Finally, extend ρ to all of $\mathcal{B}$ by setting

$$\rho(X) = \sum_{\substack{A \in \mathrm{At}(\mathcal{B}) \\ A \subseteq X}} \rho(A).$$

We verify that ρ so defined satisfies *(i)*–*(iii)*.

(i) We have that $\rho(\Omega) = c_0 \cdot \frac{\eta}{c_0} + \sum_{C \in \mathcal{C}_+} \nu(C) - c_+ \cdot \frac{\eta}{c_+} = 1$. (We understand the first term to be 0 if $c_0 = 0$.) Regularity follows from the fact that every atom gets positive measure: $\frac{\eta}{c_0} > 0$ when $c_0 > 0$ and $\nu(C) - \frac{\eta}{c_+} \geq \nu(C) - \varepsilon > 0$ for $C \in \mathcal{C}_+$.

(ii) Let $X \in \mathcal{B} \cap \mathcal{F}$ be a union of p members of $\mathcal{C}_0$ and q members of $\mathcal{C}_+$. The finite additivity of ν and ρ yields

$$\rho(X) - \nu(X) = \frac{p}{c_0}\eta - \frac{q}{c_+}\eta$$

(again with the understanding that $\frac{p}{c_0}$ is 0 if $c_0 = 0$). Both terms lie between 0 and η, so $|\rho(X) - \nu(X)| \leq \eta \leq \varepsilon < \delta$.

[2]The members of $\mathcal{C}$ are the Y_I^*'s of [3].

(iii) Suppose we have $\{x\} \in \mathcal{B} \cap \mathcal{F}$ and $\nu(\{x\}) = 0$. Then $\{x\}$ is an atom of $\mathcal{B} \cap \mathcal{F}$ and so a member of $\mathcal{C}$, indeed of $\mathcal{C}_0$ (so in particular $c_0 > 0$). Since $\{x\}$ is the only atom of $\mathcal{B}$ which it contains, $n_{\{x\}} = 1$, whence $\rho(\{x\}) = \frac{\eta}{c_0}$, which is clearly independent of x. □

Lemma 1 is not quite strong enough for our purposes, since within our finite set Γ_0 there may also be statements of the forms *(e)* and *(f)* above, and there is no guarantee that the ρ constructed above will satisfy them as well. To take care of these, we will "average" over translates of a measure constructed using Lemma 1. This is the content of:

Lemma 2. *Let $\mathcal{X}$ be a finite family of subsets of Ω and $\mathcal{B}_0$ be the Boolean algebra it generates. Given any $r_1, \ldots, r_m \in \Omega$, $N \geq 1$ and "tolerance" $\delta > 0$, there is a finitely additive $\rho\colon \mathcal{B}_0 \to \mathbb{R}$ such that*

(i) $\rho(\Omega) = 1$ *and ρ is regular;*

(ii) $|\rho(X) - \lambda(X)| < \delta$ *for every $X \in \mathcal{B}_0 \cap \mathcal{L}$;*

(iii) $|\rho(X + r_j) - \rho(X)| \leq \frac{1}{N}$ *whenever $1 \leq j \leq m$ and $X, X + r_j \in \mathcal{B}_0$; and*

(iv) $\rho(\{x\}) = \rho(\{y\})$ *whenever $\{x\}, \{y\} \in \mathcal{B}_0$.*

Proof. Given translations $r_1, \ldots, r_m \in \Omega$ and $\mathbf{k} = (k_1, \ldots, k_m) \in \mathbb{Z}^m$, put $s_{\mathbf{k}} = k_1 r_1 + \cdots + k_m r_m \in \Omega$ (we work with addition mod 1, as always). Let e_j be the j-th unit vector, so that $s_{\mathbf{k}} + r_j = s_{\mathbf{k}+e_j}$, and for $N \geq 1$ write Q_N for $\{0, 1, \ldots, N-1\}^m$.

Now, let $\mathcal{B}$ be the finite algebra generated by $\{X + s_{\mathbf{k}} : X \in \mathcal{X},\ \mathbf{k} \in Q_N\}$. Because translation by any s is a Boolean automorphism of $\mathcal{P}(\Omega)$, $X + s_{\mathbf{k}} \in \mathcal{B}$ for all $X \in \mathcal{B}_0$, $\mathbf{k} \in Q_N$. We can thus apply Lemma 1 with $\mathcal{F} = \mathcal{L}$ and $\nu = \lambda$ (and δ as given), thereby obtaining a finitely additive ρ_0. In this way we get a ρ defined on each $X \in \mathcal{B}_0$ by setting

$$\rho(X) = \frac{1}{N^m} \sum_{\mathbf{k} \in Q_N} \rho_0(X + s_{\mathbf{k}}).$$

If X is the disjoint union of X_1 and X_2 then $X + s_{\mathbf{k}}$ is the disjoint union of $X_1 + s_{\mathbf{k}}$ and $X_2 + s_{\mathbf{k}}$, so the finite additivity of ρ follows from that of ρ_0. We again verify *(i)–(iv)*.

(i) Since $\Omega + s_{\mathbf{k}} = \Omega$ for every $\mathbf{k} \in Q_N$, we have that $\rho(\Omega) = 1$. Also, since $X + s_{\mathbf{k}}$ is nonempty whenever X is, the regularity of ρ follows from that of ρ_0.

(ii) Lebesgue measure is itself translation-invariant: $\lambda(X + s_{\mathbf{k}}) = \lambda(X)$ for every $X \in \mathcal{L}$, and so it follows that each term $\rho_0(X + s_{\mathbf{k}})$ lies within δ of $\lambda(X)$. Therefore so does their average.

(iii) Since $(X + r_j) + s_{\mathbf{k}} = X + s_{\mathbf{k}+e_j}$, we have that

$$\rho(X + r_j) = \frac{1}{N^m} \sum_{\mathbf{k} \in Q_N + e_j} \rho_0(X + s_{\mathbf{k}}),$$

and so, writing $F^+ = (Q_N + e_j) \setminus Q_N$ and $F^- = Q_N \setminus (Q_N + e_j)$,

$$\rho(X + r_j) - \rho(X) = \frac{1}{N^m}\Big(\sum_{\mathbf{k} \in F^+} \rho_0(X + s_{\mathbf{k}}) - \sum_{\mathbf{k} \in F^-} \rho_0(X + s_{\mathbf{k}})\Big).$$

Notice that each index set is a face of one of the boxes $Q_N + e_j$ and Q_N (namely the faces $k_j = N$ and $k_j = 0$ respectively), and so has N^{m-1} elements. Thus both sums lie between 0 and N^{m-1}, and their difference has magnitude $\leq N^{m-1}$. Therefore $|\rho(X + r_j) - \rho(X)| \leq \frac{N^{m-1}}{N^m} = \frac{1}{N}$.

(iv) This follows from *(iii)* of Lemma 1. □

And now we are home free. For let $\mathcal{X}$ be the set of $X \subseteq \Omega$ such that $\dot{\mu}(X)$ occurs in our finite $\Gamma_0 \subseteq \Gamma$, let $r_1, \ldots, r_m$ be the translations occurring in the instances of *(f)* contained in Γ_0, and let N exceed every n occurring in the instances of *(e)* and *(f)*. We then simply apply Lemma 2 with $\mathcal{X}$, $r_1, \ldots, r_m$, N and $\delta = \frac{1}{N}$, and interpret $\dot{\mu}(X)$ as $\rho(X)$ for each $X \in \mathcal{X}$ (the remaining constants $\dot{\mu}(Y)$ can be interpreted arbitrarily). Thus Γ is finitely satisfiable, as required.

Observation 3. We've used two features of Ω in our argument. The first is that $\mathcal{L}$ is a translation-invariant algebra of subsets of Ω which contains every singleton, and λ is translation-invariant and assigns each singleton measure 0. The second is that the group $\mathbb{Z}^m$, acting on Ω by $\mathbf{k} \cdot x = x + s_{\mathbf{k}}$, has Følner sets (see [2]): in this case, the boxes Q_N, which satisfy $|(Q_N + e_j) \triangle Q_N| = 2N^{m-1} = \frac{2}{N}|Q_N|$. Our result thus generalises in the following nice way. Let G be a group acting on a set Ω, and suppose we have a G-invariant algebra $\mathcal{F}$ of subsets of Ω, as well as $\nu \colon \mathcal{F} \to [0, 1]$ finitely additive and G-invariant, with $\nu(\Omega) = 1$. If G is amenable as a discrete group, then there are ${}^*\mathbb{R} \succ \mathbb{R}$ and a regular, finitely additive $\mu \colon \mathcal{P}(\Omega) \to {}^*\mathbb{R}$ with $\mu(\Omega) = 1$, such that $\mu(X) \simeq \nu(X)$ for every $X \in \mathcal{F}$ and $\mu(gX) \simeq \mu(X)$ for every $X \subseteq \Omega$, $g \in G$; if, in addition, $\mathcal{F}$ contains every singleton and ν vanishes on them, μ can also be taken to be uniform on points. [In the proof of Lemma 2 we replace Q_N by a finite nonempty $S \subseteq G$ with $|Sg_j \triangle S| \leq \frac{2}{N}|S|$ for each of the finitely many g_j in question, and set $\rho(X) = \frac{1}{|S|}\sum_{s \in S} \rho_0(sX)$.]

References

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[2] E. Følner (1955), On groups with full Banach mean value, *Mathematica Scandinavica* 3: 243–254.

[3] T. Hofweber and R. Schindler (2016), Hyperreal-valued probability measures approximating a real-valued measure, *Notre Dame Journal of Formal Logic* 57(3): 369–374.